

Technology Stack

Date	30 September 2026
Team ID	6ab39579211fa4fd5d28901b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

EduGenie uses a lightweight web stack with a FastAPI backend, HTML/CSS/JavaScript frontend, Google Gemini API for core generative-learning features, and a local LaMini-Flan-T5-783M model for concept explanation.

The selected technologies support modular development, maintainability, and local execution.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side Web UI	HTML, CSS, JavaScript	<i>Provides the responsive web interface and user interaction for the five EduGenie learning features.</i>

2	Backend / Server-Side <i>Python runtime</i>	<i>Python + FastAPI</i>	<i>Provides API endpoints, input validation, routing, and integration with the AI feature modules.</i>
3	Database / Data Storage <i>No persistent storage</i>	<i>No persistent database</i>	<i>The current MVP does not require persistent user or application data storage.</i>
4	Cloud / Hosting / Deployment <i>Local development</i>	<i>Local Uvicorn deployment</i>	<i>Runs the FastAPI application locally for development, testing, and demonstration.</i>
5	Version Control & CI/CD	<i>Git + GitHub</i>	<i>Supports source-code version control, collaboration, documentation, and project submission.</i>

	<i>GitHub repository</i>		
6	Third-Party APIs / Other Tools <i>Google Gemini API</i>	<i>Gemini API + LaMini- Flan-T5- 783M</i>	<i>Gemini powers Q&A, quiz generation, summarization, and learning paths; LaMini provides local concept explanations.</i>